

CHAITANYA AJAY BHUJBAL

Frontend Developer • Cloud & Distributed Systems Enthusiast

Pune, Maharashtra, India | chaitanyabhujbal809@gmail.com | [linkedin.com/in/chaitanya-bhujbal-aba168332](https://www.linkedin.com/in/chaitanya-bhujbal-aba168332) | github.com/chaitanyabhujbal912006-afk | chaitanya-bhujbal-portfolio.vercel.app

SUMMARY

Third-year Computer Science & Engineering student and backend developer focused on cloud infrastructure, distributed systems, and AI-driven healthcare technology. Competes in national-level hackathons building multi-agent AI and systems-security projects, ships production-oriented tools (a published PyPI security scanner, a self-hosted RAG platform), and pairs technical work with a strong design sensibility — leading design for a student tech community and creating visual content for college-wide events.

TECHNICAL SKILLS

- **Languages:** Java, C++, Python, JavaScript, TypeScript
- **Backend & Frameworks:** Node.js, Express.js, Next.js, React, REST & GraphQL APIs
- **Cloud & DevOps:** AWS, Docker, Kubernetes (learning), Git & GitHub Actions, CI/CD pipelines
- **Databases:** PostgreSQL, MongoDB, Redis
- **Other:** System Design, Retrieval-Augmented Generation (RAG), Multi-Agent AI Systems, Tailwind CSS

KEY PROJECTS

ReachGuard — Reachability-Aware Dependency Vulnerability Scanner (*Python, PyPI, PyCG, OSV.dev*)

- Published a zero-cost, local-first CLI to PyPI performing static call-graph reachability analysis, cutting false-positive CVE alerts by 60–80% versus tools like Dependabot/Snyk; integrates as a GitHub Action, backed by a 35-test suite.

Nexa — Zero-Cost Retrieval-Augmented Generation (RAG) Knowledge Base (*Python*)

- Built a RAG system that ingests PDFs, Excel sheets, and emails into a single searchable knowledge base, answering natural-language questions with citations at zero infrastructure cost.

EdiPro / EDI Healthcare — HIPAA-Compliant Healthcare EDI Parser (*Python*)

- Built an enterprise-grade Electronic Data Interchange parser, validator, and dashboard transforming clinical/financial EDI transactions into structured, actionable data.

SnapStudio AI — AI-Powered Photo Editing Platform (*Next.js, Python, Vercel, Kaggle GPU*)

- Shipped a full-stack app pairing a Next.js frontend (Vercel) with a GPU-backed inference service (Kaggle) for instant auto-enhance, background swap, and object removal.

Navonmesh 2026 — Official Hackathon Website (*Web*)

- Developed the official website for Navonmesh 2026, a national-level 24-hour hackathon, handling event information, schedules, and registrations.

HACKATHONS

OmniKavach — Agentic ICU Diagnostic Assistant | *Ignisia 2026, National-Level AI Hackathon*

- Architected a multi-agent clinical decision-support system integrating the MIMIC-III dataset to flag early warning signs of critical conditions like sepsis, powered by a “Chief Synthesis Agent” and “Temporal Lab Mapper” for real-time patient data analysis.
- Built the frontend (Vite, React, Tailwind CSS) connected to Python/FastAPI backend agent orchestration; owned system architecture and UI within a cross-functional team.

Accessibility AI Assistant | *Pragyantra 2026 — Healthcare Domain, PES Modern College of Engineering*

- Built an AI-powered assistant to streamline applying for government benefit schemes, with an intuitive, accessible UI for users with disabilities.
- Delivered end-to-end full-stack development and system integration within a strict 8-hour build sprint.

Sentinel (Kernel Guard) — Real-Time Zero-Day Ransomware Defense | *TechVortex 3.0 — Top 50 Finalist (150+ International Teams)*

- Architected a real-time, behavior-based ransomware mitigation system using eBPF kernel hooks to detect zero-day encryption attacks at low latency with minimal CPU overhead.
- Built a high-throughput Node.js telemetry pipeline feeding a Groq-accelerated LLaMA inference engine to flag anomalous file access and auto-terminate malicious processes before encryption completed.

DESIGN & CREATIVE WORK

- **Design Head & Core Member**, AWS Student Builder Club — lead visual/creative direction as a core member, designing graphics and collateral for club initiatives and events.
- Manage and design content for official college social media pages; designed posters and promotional artwork for 6+ college events, including the Tech Fest of the School of Engineering & Technology.
- Practices canvas painting as a personal creative pursuit.

EDUCATION

B.Tech in Computer Science & Engineering — DES Pune University (DESPU), Pune, Maharashtra

Currently in Third Year | Focus: Distributed Systems, Cloud Infrastructure, Backend Architecture, Healthcare Technology